HOW WE TESTED

We believe that a motherboard's performance is completely a function of its components. The higher specced the components, the better the performance that you'll get. For the same chipset, the variation in performance is not very evident from our scores. The difference is within the margin of error. This can be easily understood by the fact that benchmark scores from a ₹8,000 board are quite similar to the benchmark scores from a ₹30,000 motherboard, give or take five percent.

So it ultimately boils down to the feature set you're getting with the motherboard, the build quality, ease of overclocking and other factors. Keeping this in mind, we decided to not give any weightage to the performance scores. We have run the benchmarks, and the scores are shared in the tables on the following pages. In our final evaluation though, we've not considered these scores as they don't give you an estimation of the motherboard's value.

Our Test Rig comprised of the following components:

For Intel Z68 boards
Processor: Intel Core i7-2600K
HDD: WD Velociraptor 600 GB
RAM: 2x2GB Kingston HyperX RAM
Graphics Card: ZOTAC GTX 560
Power Supply: Cooler Master GX 650W
OS: Windows 7 Ultimate x64

For AMD Boards
Processor: AMD A8-3850
HDD: WD Velociraptor 600 GB
RAM: 2x2GB Kingston HyperX RAM
Graphics Card: Integrated GPU AMD HD6550D
Power Supply: Cooler Master GX 650W
OS: Windows 7 Ultimate x64

The layout of the board was scored, we checked if there was enough spacing around the CPU region to comfortably attach the CPU cooler and which direction the SATA ports were pointing (those pointing up got a lesser score). Spacing between the RAM slots and the graphics card was checked. We also noted the spacing between PCIe and PCI slots. If a dual-slot card blocked the next PCIe slot, marks were deducted.

For build quality we noted the sturdiness of the heat sinks around the VRM (Voltage Regulator Module) regions and the southbridge. Motherboards lacking a heatsink around the VRM region got a lesser score. Boards with all solid state capacitors got a higher score. Back panel I/O ports were noted and also the option to expand USB ports. Fan connectors and SATA connections were noted.

Overclocking features such as utilities and on-board chip additions were taken into consideration. Boards having more options to overclock and tweak got a higher score. Also, the BIOS user experience was noted. Boards which didn't have a UEFI BIOS got a lower score since that's the future.

We did a thorough checking and testing (wherever applicable) of the feature set offered by a motherboard and based on the price, have decided on the winners.

digital audio processor. It also houses on-board amplifiers to drive audio. On-board implementation of a good sound processor is impressive for most users, but hardcore audiophiles will be disappointed and will invest in a dedicated sound card anyway. There's a dedicated overclock button on the back I/O panel to overclock the processor on the fly without having to reboot the system. Display ports are naturally absent. And needless to say, it's high time Gigabyte revamped the BIOS, Touch BIOS is a gimmick and is nothing like the experience on a UEFI BIOS.

The ASUS Maximus IV Extreme Z belongs to the Republic of Gamers (RoG) series. Build Quality is top-notch with attractive crimson-black heat sinks with multiple fins around the CPU. The RoG logo glows red when the system is powered on. Eight fan-connectors on-board will ensure sufficient cooling for your system.

The Maximus IV Extreme houses an Nvidia NF200 chip which controls second and fourth PCIe16x slot, while the processor's integrated controller controls the first and third. What this means is that you can run two

Gigabyte Z68XP-UD3-iSSD

cards in SLI at x16/x16 speed (slot 2 and 4), but there isn't much of a difference between the two configurations as latency from the NF200 chip has to be taken into account. So, this shouldn't be one of the reasons for you to consider this board. For those planning on a three-way SLI setup, NF200 does help.